31338 Network Servers

Lab 2b: CentOS and Windows System Updates

Aims

1. Update your system

It is vital to keep your operating systems up to date with security patches and fixes. However, it is also important to ensure that your operating system is not affected by unnecessary upgrades.

Task 1: CentOS Updates

To install updates, your server needs to be connected to the internet. In the Linux virtual machine supplied, check the status of NetworkManager service:

```
systemctl status NetworkManager.service
```

Note the commands and steps in your learning journal.

Once that's done, if you click on the top right of the screen, turn on *Ethernet (ens160)*.

By default, CentOS uses the Gnome Software app to monitor and update system software on your Linux configuration. Along the top of the window there should be a tab called "Updates". There is a "Refresh" icon that has a circular arrow. Press this to check for new updates.

Observe the list of software to update, if any. You should make a judgement on which software to install (they are ALL selected by default, but is this a good idea?)

Be EXTREMELY careful of any changes to your Kernel or GLIB - these changes can cause operational issues as you are installing.

You can use the yum command to update packages. This is often a safer way of doing updates.

Yum commands include:

```
yum update
yum search <package name>
yum install <package name>
yum update
yum remove <package name>
```

The yum automatically determines the dependencies and can warn you. You can also use the various options of yum to only load security or advisory fixes. Most of times, we also use *dnf* to manage the software packages. The usage is like yum, e.g.,

```
dnf install cups
```

Task 3: Windows Server Updates

Windows manages updates mainly through GUI. However, it is not recommended to automatically perform updates on a running server. Can you think of any reasons why?

Within Server Manager, on your *Local Server* the dashboard should show you some information about when updates were last installed and the update policy.

To run updates, you can use *Settings* -> *Windows Update*. You can check the updates and install them.

Generally, Windows Server updates are usually well tested and can be installed. Be careful of optional updates such as. NET framework etc.

You can also check for and install updates in Windows from the command line. While you can do it from the Command Prompt, Windows PowerShell provides a module for working with Windows update from PowerShell. To install the module, type:

```
Install-Module PSWindowsUpdate
```

It will probably prompt you / warn you about installing other dependent software - choose 'Yes' if asked.

Once installed, you can use PowerShell commands (cmdlets) like:

```
Get-WindowsUpdate
Get-WULastResults
Get-WUHistory
Get-WURebootStatus
```

Try some of these and record notes in your learning journal. Search online for help with the "PSWindowsUpdate" module.

Run the updates now (GUI or command-line), but beware, this can take a while to download and install.